

Scope: 270-device fleet · 4-month season · 8 regattas · 32 tracking days. Two tiers: **ops / tracking only** (fleet + platform + data) vs + **virtual livestream** (2D race viewer / on-air graphics — no on-site cameras). CrewSight season quote is **ops / tracking** at \$46,500 — vMix / AHD overlay feeds **supported**, regatta production not included. Optional CrewSight **virtual livestream** add-on: \$750/regatta day × 32 days = \$24,000. GeoRacing per Trimaran preliminary budget: tracker **purchase** €140/unit; Safety €150/day · Fan Experience €300/day + API €200/day for public/broadcast. SIMs **not included** in GeoRacing hardware quote — comparison adds \$5/SIM/mo for parity. EUR → NZD @ 1.82.

Season cost summary — 270 devices (ex GST)

Solution	Ops / tracking only	+ Virtual livestream	Notes
CrewSight	\$46,500 \$53,475 incl	\$70,500 \$81,075 incl	Ops / tracking in base quote. Livestream overlay feeds supported. Virtual livestream: +\$750/day × 32 days. No on-site cameras.
IQnexus LoRaWAN	\$67,000	\$67,000	No livestream path — tracking portal only
GeoRacing (Trimaran budget)	\$84,388 €43,400 + SIM	\$104,772 €54,600 + SIM	Ops: Safety package €150/day. Virtual: Fan Experience + API (€300 + €200/day × 32 days). TV GFX rental optional extra.

Per device (ops tier, incl. modelled SIM): CrewSight \$172 · LoRa \$248 · GeoRacing \$313 (€161 hardware/licensing + SIM)

CrewSight virtual livestream add-on: \$750 × 32 regatta days = \$24,000 ex GST. GeoRacing virtual tier upgrade: Fan Experience + API = €11,200 (\$20,384 NZD) over Safety package for 32 days.

GeoRacing preliminary budget — EUR → NZD

Trimaran email quote: hardware purchase + per-event licensing (excl. tax/shipping). 270 devices scaled from vendor example (120 × €140). SIMs modelled at \$5/device/mo × 4 mo (not in GeoRacing quote). Conversion: 1 EUR = 1.82 NZD.

Tier	Includes	EUR (ex tax)	NZD + SIM
Ops / tracking only	270× tracker purchase · €800 training · Safety €150/day × 32 days	€43,400	\$84,388
+ Virtual livestream	Above with Fan Experience €300/day + API €200/day (public viewer + broadcast partner) — no on-site cameras	€54,600	\$104,772

GeoRacing line items (270 devices · 32 tracking days)

Item	Ops (EUR)	+ Virtual (EUR)
GPS trackers (270 × €140, purchase)	€37,800	€37,800
Initial training (1-day webinar)	€800	€800
Safety package (private monitoring)	€4,800 32 × €150	—
Fan Experience (public iframe / app / leaderboard)	—	€9,600 32 × €300
GEORACING API (livestream partner graphics)	—	€6,400 32 × €200
Subtotal (ex tax/shipping/SIM)	€43,400	€54,600
Cellular SIMs (modelled — not in GeoRacing quote)	\$5,400	\$5,400
NZD total (incl. modelled SIM)	\$84,388	\$104,772

Optional extras not in totals: remote support €450/day · on-site €550/day · TV graphics rental €500/day + setup €1,200–€3,000.

GeoRacing & CrewSight — virtual livestream line item

Item	CrewSight (NZD)	GeoRacing (NZD)
Fleet / platform (ops / tracking)	\$46,500	\$84,388

Item	CrewSight (NZD)	GeoRacing (NZD)
Virtual livestream licensing	\$24,000 32 days × \$750	\$20,384 €11,200 (Fan + API – Safety)
Total with virtual livestream	\$70,500	\$104,772

Line-item comparison — ops / tracking (270 devices · 4 mo)

Cost element	CrewSight	LoRaWAN	GeoRacing (NZD)
End devices	\$35,100 purchase	\$53,730 purchase	\$68,796 purchase €140/unit
Gateways / infrastructure	\$0	\$9,800 (7×)	\$0
Platform / event licensing	\$6,000	\$2,970	\$10,192 training + Safety 32d
Cellular / IoT data	\$5,400	\$0*	\$5,400 not in GeoRacing quote
Livestream data / vMix feeds	Supported	Not included	API tier (+virtual)
Virtual livestream	Optional +\$24,000 \$750/day	—	Optional +\$20,384 Fan + API – Safety
On-site cameras	<i>Not included — neither CrewSight nor GeoRacing virtual tier uses on-site cameras</i>		
Boat mounting (rowing)	\$50/boat optional	Not offered	Not offered
Total ex GST	\$46,500	\$67,000	\$84,388

Device & hardware comparison

Side-by-side tracker/handset characteristics for ops / tracking. Season costs in earlier tables include platform, data, and gateways where applicable.

Attribute	CrewSight	IQnexus LoRaWAN	GeoRacing
Device type	Android smartphone (One NZ Smart M26)	Dedicated LoRa GPS tracker	Dedicated LTE GPS tracker (purchase)
Device cost (270 fleet)	\$35,100 purchase \$130/unit · KRI owns	\$53,730 purchase \$199/unit · KRI owns	\$68,796 purchase €140/unit · KRI owns · excl. tax/shipping
Extra infrastructure	None (cellular)	\$9,800 — 7× outdoor gateway @ \$1,400 + \$500 setup · WAN backhaul per site	None — cellular trackers
Connectivity / range	Cellular LTE — national coverage; not line-of-sight limited	LoRaWAN to gateway — line-of-sight ~2–5 km per outdoor gateway; plan coverage for full 2 km course + return lanes	LTE Cat M1 + 2G fallback — cellular; not line-of-sight limited
GPS update rate	1 Hz on water (field-validated); configurable 1 s–30 s+ Mixed regatta profile: 1 s / 5 s / 30 s	10 s assumed (~0.10 Hz) — confirm with IQnexus	<i>Not stated in vendor materials — OTA configurable; confirm with Trimaran</i>
GPS accuracy	3.3 m median (field test 14 Jun 2026)	<i>Not quoted — confirm with vendor</i>	1 m (vendor spec)
Battery / runtime	5,000 mAh · ~32 h @ 1 Hz continuous ~25% for 8 h regatta day — overnight charge	<i>Not quoted — typically long-life replaceable cell at low duty cycle; confirm</i>	2600 mAh · 14–70 h autonomy (vendor spec; rate-dependent)
Weight / size	Smartphone (~5000 mAh handset)	<i>Compact tracker — confirm model with IQnexus</i>	93 g · 39.9 × 26.7 × 77.9 mm
Capsize / safety alert	Automatic hull & crew event detection (rowing-tuned)	Generic motion — not rowing-specific	Manual emergency button only
Offline buffer	App queues uploads; cellular backfill when coverage returns	LoRa store-and-forward via gateway	Up to 10,000 messages (vendor spec)

Attribute	CrewSight	IQnexus LoRaWAN	GeoRacing
Rowing boat mounting	Optional \$50/boat	Not offered	Not offered
Livestream / vMix data feed	Supported — AHD overlay APIs	Not available	Fan Experience + API (€300 + €200/day) — optional TV GFX rental extra
Cellular data (season)	\$5,400 (\$5/SIM/mo × 270)	\$0 per device (gateway WAN only)	<i>Not included in tracker price — budget \$5,400 (\$5/SIM/mo × 270, same as CrewSight)</i>

Sources: CrewSight — M26 field test 14 Jun 2026 (Karāpiro). LoRaWAN — IQnexus pricing & 10 s uplink assumption; gateway range indicative. GeoRacing — Trimaran preliminary budget email + Fan Experience PDF (hardware specs); reporting rate not published.

Ops / tracking vs virtual livestream

Capability	CrewSight (ops)	CrewSight (+ virtual)	LoRaWAN	GeoRacing (ops)	GeoRacing (+ virtual)
Season cost (NZD)	\$46,500	\$70,500	\$67,000	\$84,388	\$104,772
Tracking data for livestream	Supported — vMix / AHD feeds	Supported — vMix / AHD feeds	Not available	<i>Safety (private)</i>	Fan Experience + API
Virtual livestream (2D / GFX)	<i>Optional add-on</i>	\$750/regatta day	—	<i>Safety only</i>	Fan €300/d + API €200/d
On-site cameras	<i>Not included</i>		—	<i>Not included</i>	
Capsize / hull alert	Yes — rowing-tuned		Generic motion	Emergency button	
Rowing dashboard	Karāpiro · RowIT · ops monitor		Generic IoT	Custom per event	
Boat mounting	Optional \$50/boat		Not offered	Not offered	
Device ownership	KRI owns handsets		KRI owns	KRI owns trackers (purchase)	
Handset field test (Jun 2026)	M26: 3.3 m GPS · ~32 h @ 1 Hz		<i>Not quoted</i>	93 g LTE tracker (vendor spec)	

M26 field validation (14 Jun 2026)

Field test 14 Jun 2026: 11:30–11:45 NZST drive (~10 km) at ~1 Hz. M26 median GPS accuracy **3.3 m** (p90 5.2 m) vs S21 reference 5.2 m; fix continuity 1 s max gap on M26. Track agreement median 4.2 m.

Battery 14 Jun 2026: 96% @ 06:45 NZ → 64% after 10.3 h (~3.1 %/h at 1 Hz GPS + background recording; ~32 h from full charge; app data 36 MB). An 8-hour active day at 1 Hz uses ~25% — no mid-day charge required.

Ops / tracking (CrewSight \$46,500): GPS fleet, platform, and data — vMix / AHD overlay feeds supported for livestream (regatta production quoted separately). **Virtual livestream** optional — CrewSight at \$750/regatta day (32 days = \$24,000); GeoRacing Fan Experience + API (€300 + €200/day). 2D race viewer / broadcast API; **no on-site cameras**.

CrewSight strengths

- **Capsize alert · Custom rowing dashboard · Designed for NZ regatta programme**
- **Livestream supported** — vMix / AHD overlay feeds in ops tier · optional **virtual livestream** at \$750/regatta day
- **Rowing boat mounting** optional; competitors do not supply shell mounting

Summary

Ops / tracking: CrewSight \$46,500 vs LoRa \$67,000 vs GeoRacing \$84,388 (€43,400 + modelled SIM). **+ Virtual livestream:** CrewSight \$70,500 (+\$24,000 at \$750/day) vs GeoRacing \$104,772 (€54,600 + SIM). Neither vendor includes on-site cameras. Both CrewSight and GeoRacing are purchase-based CAPEX; GeoRacing adds per-event licensing.

Optional CrewSight mounting (270 boats): +\$13,500 ex GST. LoRa gateway WAN backhaul not itemised. GeoRacing pricing from Trimaran preliminary budget email; tax/shipping/SIM configuration extra.